

Data Analytics (DA)

Module -2

Power BI Assignment 2 – DAX & Data Visualization

Assignment Task Overview

Problem Statement:

As a **Data Analyst**, you will continue working with the **E-Commerce Sales Dataset** used in the previous assignment. The dataset already contains the required tables and relationships created during **data transformation and data modeling**.

In this assignment, you will use **DAX (Data Analysis Expressions)** in Power BI to create calculated columns and measures that generate meaningful business metrics. You will then develop **interactive visualizations** to analyze sales trends, profitability, and performance indicators within the e-commerce business environment.

The focus of this assignment is on **DAX calculations, analytical thinking, and visual storytelling**, while ensuring that the underlying **data model supports accurate reporting and insights**.

Dataset Description :

You will work with the following three datasets:

- [List of orders](#)

***Attributes:** Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit*

- [Order Details](#)

***Attributes:** Order ID, Category, Sub-Category, Quantity, Amount, Profit*

- [Sales Target](#)

***Attributes:** Category, Month of Order Date, Target*

Tasks

DAX Calculations

Calculated Columns

- **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

- **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.
- **Create a Calculated Column to Categorize Sales:** Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Calculated Measures

1. **Order Count-** Create a measure that calculates the **total number of orders** in the **Order Details** table.
2. **Average Profit in Delhi-** Create a measure that calculates the **average profit for orders** placed in Delhi.
3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date**.

Data Visualization

Create the following visualizations using Power BI:

- **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.
- **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.
- **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.
- **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.
- **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison

and analysis. Also, create a multi-row card to display the minimum target for each segment.

- **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.
- **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.
- **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a treemap.
- **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.

Deliverables:

Submit a project folder containing:

1. Power BI File (.pbix)

The Power BI file should include:

- Data model created in Assignment 1
- All calculated columns
- All DAX measures
- Completed visualizations
- Proper formatting and labels

2. PDF Document

Include screenshots of:

- DAX formulas used
- Created visualizations
- Dashboard layout
- Provide a brief explanation for each visual.

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Calculated Columns	Data Modeling	Column Calculations	Beginner
DAX Measures	Data Analysis	Aggregations & Filters	Intermediate
Time Intelligence	Analytical Modeling	YTD Calculations	Intermediate
Data Visualization	Data Storytelling	Charts & Graphs	Beginner
Dashboard Design	Data Presentation	Formatting & Layout	Intermediate
Business Analysis	Analytical Thinking	Sales Performance Analysis	Intermediate

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Calculated Columns	2	Correct column formulas and logic	Minor formula errors	0.5-1
DAX Measures	3	Correct measures with proper aggregation	Partial logic issues	1-2
Visualizations	3	Correct chart types with proper fields	Incorrect chart usage	1-2
Dashboard Design & Formatting	2	Clear, readable, and well-formatted visuals	Poor formatting or missing labels	0.5-1

DA Skill Taxonomy Structure

- **Data Modeling:** Creating calculated columns and derived attributes
- **Data Analysis:** Writing DAX measures for business metrics
- **Time Intelligence:** Implementing YTD calculations
- **Data Visualization:** Creating charts and dashboards
- **Business Interpretation:** Identifying trends and insights from visual reports
- **Power BI Proficiency:** Implementing industry-standard reporting practices

Evaluation Guidelines

- Verify calculated columns and DAX formulas for correctness.
- Validate that the **data model supports the calculations**.
- Ensure appropriate visual types are selected for analysis.
- Check formatting, labeling, and readability of dashboards.
- Assign marks based on **accuracy, logic, and clarity of visual storytelling**.

Achievement Criteria & Passing Rules

- Students must obtain **partial marks in every assignment section**.
- Students must demonstrate proficiency in **DAX calculations and visualization design**.
- All visualizations must be **correctly configured and clearly labeled**.
- The submitted Power BI report should clearly communicate **business insights**.